

26 JUNE 2024

Fridgemate

Making Forgotten Food Visible Before It Becomes Waste

WHAT IT IS

Overview

FridgeMate is a phygital refrigerator companion designed to make fridge inventory easier to track and manage. It combines camera-assisted inventory tracking with a refrigerator-mounted interface to keep users informed about what food is stored, how long it has been there, and which items may need attention reducing the need to rely on memory or manually maintain an inventory.



HOW IT STARTED

Observation

One Friday evening, while deciding what to cook, I opened the fridge and found spoiled tomatoes, overripe fruits and leftovers we had completely forgotten about. Together, it was roughly **₹200–₹300 worth of food** not wasted because we didn't want it, but because we had simply lost track of it.

That was when I realized our refrigerator had become an “**electric almirah**” good at storing food, but not at helping us remember what was inside.



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Food can remain unnoticed in the refrigerator until it has already spoiled, with users often realizing it only when they encounter the item again.

— HYPOTHESIS

Research

To understand whether my initial observation reflected a wider problem, I explored household and fridge food waste through **secondary research, surveys, and user interviews**. The research helped establish the scale of the problem and understand how commonly stored food ends up being overlooked or wasted in everyday fridge use.

68M+ tonnes

food waste comes from Indian households annually

64%

linked to food stored/managed in refrigerators

58%

throws spoiled food from fridge

Key Insights

After completing the research, I analyzed the survey data and interview responses to identify recurring patterns in how users manage food inside their refrigerators. Synthesizing these findings revealed four key insights that captured the recurring gaps in visibility, awareness, and everyday tracking of stored food.

Lack of visibility and awareness

**Users often forget what is already
inside the fridge**

Users forget when the food was stored

**Spoilage is often noticed only after
obvious signs appear**

HOW MIGHT WE

“

How might we make it easier for users to keep track of food in their fridge before it is forgotten or goes bad?

Concept 01 — Food Spoilage Detection

The first concept explored a spoilage-detection device combining manual food scanning with gas sensing. Users would scan food before placing it inside the fridge, creating a digital record of stored items, while sensors inside the refrigerator would monitor gases associated with food deterioration and alert users when an item might be going bad.



Limitations

Evaluating the first concept revealed several gaps in both the user experience and system feasibility. These limitations showed that the solution was adding effort while still leaving critical gaps in how food was tracked and spoilage was detected.

- *Too much manual effort*
- *The inventory loop was incomplete*
- *Spoilage detection was difficult to attribute*
- *The solution intervened too late*

WHAT I CHANGED

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The limitations of the first concept changed how I approached the solution. Rather than building a system that depended on users to manually log food and then waited for signs of spoilage, I shifted toward maintaining an automatic record of what was inside the fridge and how long it had been there.

WHAT I DESIGNED

Concept 02 — FridgeMate

A passive fridge inventory companion

FridgeMate evolved into a **phygital inventory-management system** that automatically **keeps track of food** entering and leaving the refrigerator.



Key features

After identifying the gaps in the initial concept and revisiting the key insights from the research, the solution was refined into FridgeMate. The revised concept integrates features that directly respond to the user problems uncovered during the process.

- ✓ ***Automatic Inventory Tracking***
- ✓ ***Storage-Time Tracking***
- ✓ ***Inventory Visibility***
- ✓ ***Predictive Attention***
- ✓ ***Timely Alerts***

10:24

Tuesday, Apr 16

Your Fridge

Fresh picks and essentials

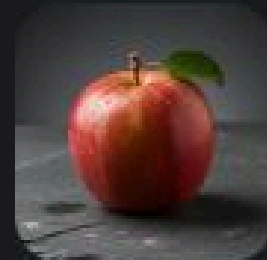
12 items

All

Dairy

Fruits

Vegetables



Apple

Since Apr 12

4 days



Eggs

Since Apr 15

1 day



Spinach

Since Apr 13

3 days



2 items have been in your fridge for over a week - use them soon!



Home



Grocery



Recipes



Add food



10:24

Tuesday, Apr 16

Kitchen

Your Fridge

Fresh picks and essentials

12 items



Inventory
Visibility

See everything
in your fridge
at a glance.



Automatic
Inventory
Tracking

Items are detected
and added to your
inventory
automatically.



Apple

Since Apr 12

4 days



Eggs

Since Apr 15

1 day



Spinach

Since Apr 13

3 days



Predictive
Attention

Get insights on
what may spoil
soon.



2 items have been in your fridge for over a week - use them soon!



Timely Alerts

Get notified about items
that need your attention.



Home



Grocery



Recipes



Add food